MSG No. 138.519 $P4_2/ncm$ [Type I, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[1,6,7,8]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[2,3,4,5]`
3	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[9,14,15,16]`
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[10,11,12,13]`

Table 2: Wyckoff site: `4b`, site symmetry: `-4..`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[1,6,10,11]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,3,9,14]`
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[4,5,15,16]`
4	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[7,8,12,13]`

Table 3: Wyckoff site: `4c`, site symmetry: `..2/m`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,8,9,16]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[2,4,10,12]`
3	$\left[0, \frac{1}{2}, 0\right]$	`[3,5,11,13]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[6,7,14,15]`

Table 4: Wyckoff site: `4d`, site symmetry: `..2/m`

No.	position	mapping
1	$[0, 0, 0]$	`[1,8,9,16]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[2,4,10,12]`
3	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[3,5,11,13]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[6,7,14,15]`

Table 5: Wyckoff site: `4e`, site symmetry: `2.mm`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,6,15,16]`
2	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[2,3,12,13]`
3	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[4,5,10,11]`
4	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[7,8,9,14]`

Table 6: Wyckoff site: `8f`, site symmetry: `2..`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[1,6]`
2	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[2,3]`
3	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[4,5]`
4	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[7,8]`
5	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	`[9,14]`
6	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[10,11]`
7	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[12,13]`
8	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[15,16]`

Table 7: Wyckoff site: `8g`, site symmetry: `..2`

No.	position	mapping
1	$\left[x,\ -x,\ \frac{1}{2}\right]$	`[1,8]`
2	$\left[x+\frac{1}{2},\ x,\ 0\right]$	`[2,4]`
3	$\left[-x,\ \frac{1}{2}-x,\ 0\right]$	`[3,5]`
4	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[6,7]`
5	$\left[-x,\ x,\ \frac{1}{2}\right]$	`[9,16]`
6	$\left[\frac{1}{2}-x,\ -x,\ 0\right]$	`[10,12]`
7	$\left[x,\ x+\frac{1}{2},\ 0\right]$	`[11,13]`
8	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[14,15]`

Table 8: Wyckoff site: `8h`, site symmetry: `..2`

No.	position	mapping
1	$[x,\ -x,\ 0]$	`[1,8]`
2	$\left[x+\frac{1}{2},\ x,\ \frac{1}{2}\right]$	`[2,4]`
3	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	`[3,5]`
4	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ 0\right]$	`[6,7]`
5	$[-x,\ x,\ 0]$	`[9,16]`
6	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}\right]$	`[10,12]`
7	$\left[x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	`[11,13]`
8	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ 0\right]$	`[14,15]`

Table 9: Wyckoff site: `8i`, site symmetry: `..m`

No.	position	mapping
1	$[x,\ x,\ z]$	`[1,16]`
2	$\left[\frac{1}{2}-x,\ x,\ z+\frac{1}{2}\right]$	`[2,12]`
3	$\left[x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[3,13]`
4	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$	`[4,10]`
5	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[5,11]`
6	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ z\right]$	`[6,15]`
7	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ -z\right]$	`[7,14]`

continued ...

Table 9

No.	position	mapping
8	$[-x,\, -x,\, -z]$	`[8,9]`

Table 10: Wyckoff site: `16j`, site symmetry: `1`

No.	position	mapping
1	$[x,\, y,\, z]$	`[1]`
2	$\left[\frac{1}{2}-y,\, x,\, z+\frac{1}{2}\right]$	`[2]`
3	$\left[y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	`[3]`
4	$\left[x+\frac{1}{2},\, -y,\, \frac{1}{2}-z\right]$	`[4]`
5	$\left[-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[5]`
6	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	`[6]`
7	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, -z\right]$	`[7]`
8	$[-y,\, -x,\, -z]$	`[8]`
9	$[-x,\, -y,\, -z]$	`[9]`
10	$\left[y+\frac{1}{2},\, -x,\, \frac{1}{2}-z\right]$	`[10]`
11	$\left[-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	`[11]`
12	$\left[\frac{1}{2}-x,\, y,\, z+\frac{1}{2}\right]$	`[12]`
13	$\left[x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	`[13]`
14	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	`[14]`
15	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z\right]$	`[15]`
16	$[y,\, x,\, z]$	`[16]`